Cognifyz Technologies

Restaurant Data Analysis Report

Data Science Internship – Levels 1, 2 & 3

Dataset: 9,551 restaurants × 21 columns across 15 countries

Tools: Python, pandas, matplotlib, seaborn, scikit-learn

Executive Summary

- The dataset has **9,551 rows and 21 columns**; only *Cuisines* has missing values (9 rows).
- **2,148 restaurants (22.5%) are unrated** (rating 0), which skews averages.
- India holds **90.6%** of restaurants; New Delhi is the top city.
- Only **12.1%** offer table booking and **25.7%** offer online delivery.
- Higher price range goes with higher ratings (avg 2.0 in range 1 vs 3.8 in range 4).
- Best model on rated restaurants: Gradient Boosting, $R^2 \approx 0.65$, RMSE ≈ 0.33.
- Votes, price range, number of cuisines, delivery and table booking are the main drivers of rating.

Level 1

Task 1 – Data Exploration & Preprocessing

Shape: 9,551 rows, 21 columns. **Duplicates:** 0.

Missing values: Cuisines (9). Filled with "Not Available".

Type conversion: Yes/No columns converted to categorical; Country Code mapped to country names.

Target distribution (Rating text)

Class	Count	Percent
Average	3737	39.13
Not rated	2148	22.49
Good	2100	21.99
Very Good	1079	11.30
Excellent	301	3.15
Poor	186	1.95

Class imbalance: the largest class (Average) is 20.1× the smallest (Poor). 'Excellent' and 'Poor' are rare, so any classification model would need balancing; regression is more suitable.

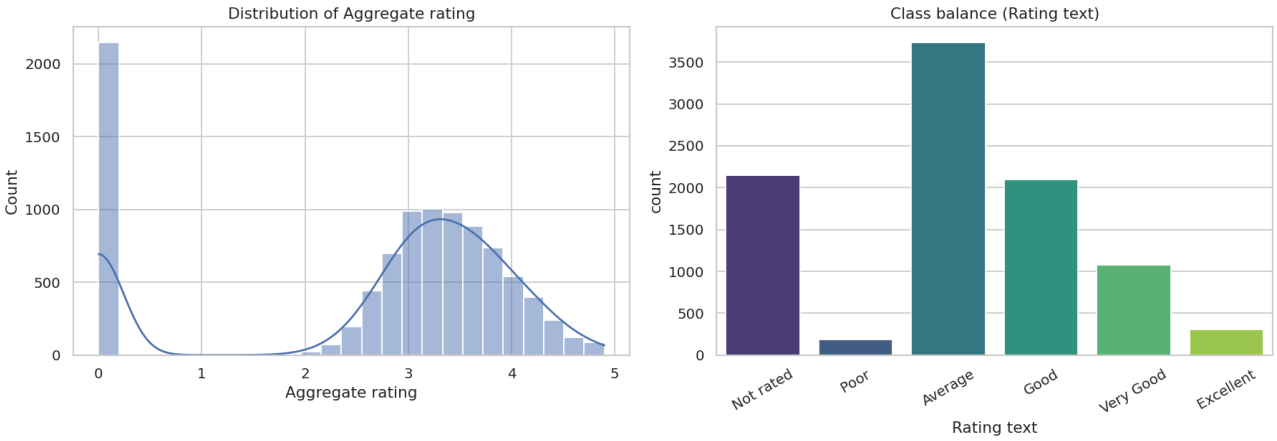


Figure 1 - Distribution of Aggregate rating and rating classes

Task 2 – Descriptive Analysis

index	mean	median	std	min	max
Average Cost for two	1199.21	400.00	16121.18	0.00	800000.00
Price range	1.80	2.00	0.91	1.00	4.00
Aggregate rating	2.67	3.20	1.52	0.00	4.90
Votes	156.91	31.00	430.17	0.00	10934.00

Top cuisines: North Indian (3960), Chinese (2735), Fast Food (1986), Mughlai (995), Italian (764).

Top cities: New Delhi (5473), Gurgaon (1118), Noida (1080), Faridabad (251), Ghaziabad (25).

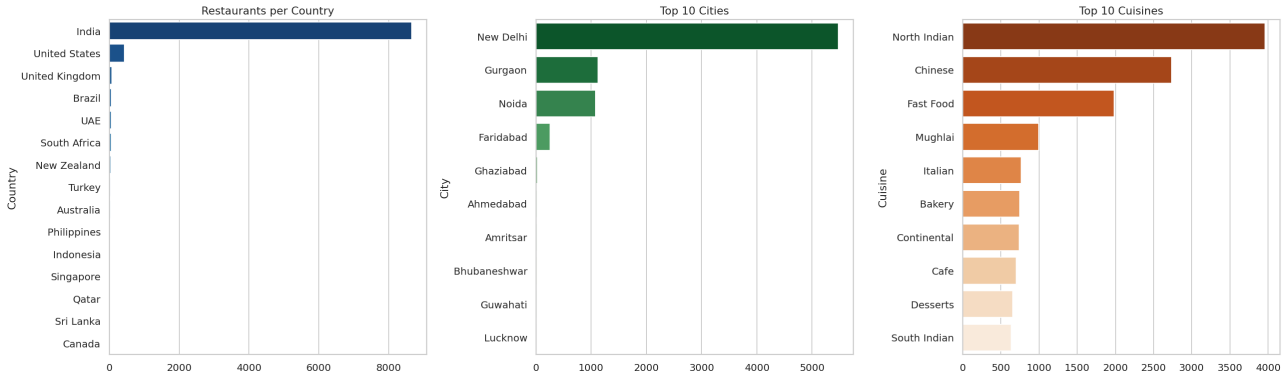


Figure 2 - Restaurants per country, top cities and top cuisines

Task 3 – Geospatial Analysis

9,054 restaurants have valid coordinates; 497 had 0,0 placeholders and were excluded from maps.

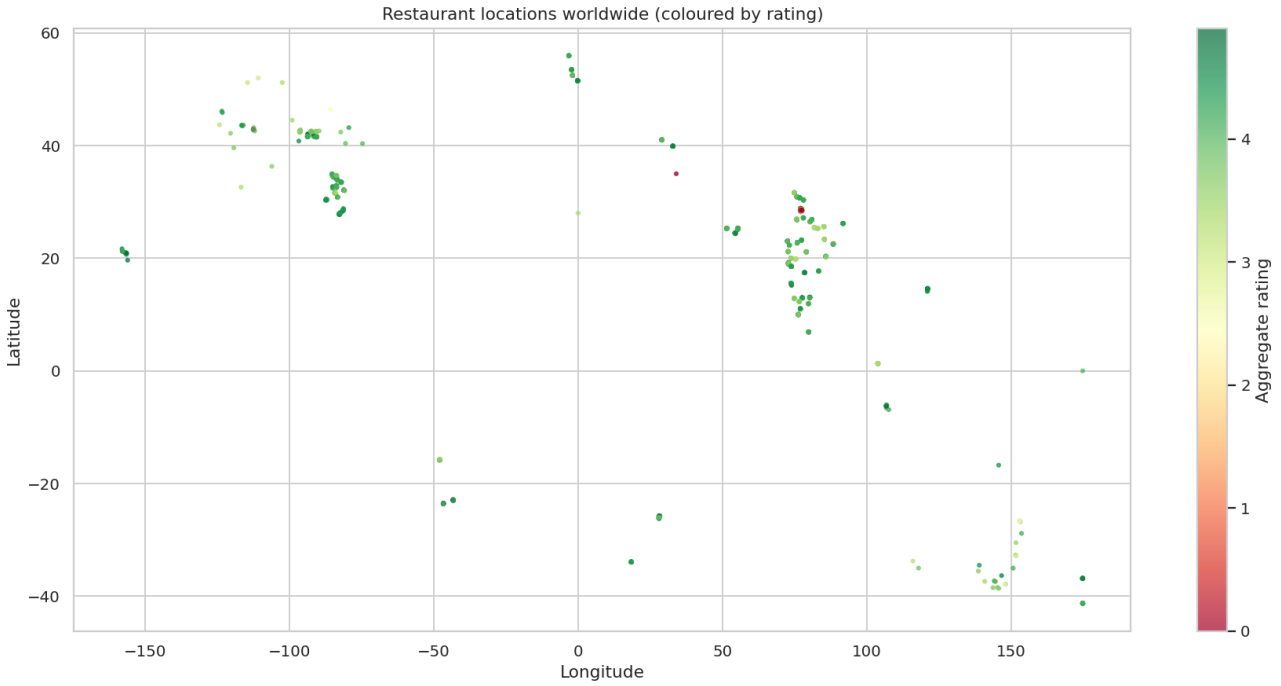


Figure 3 - Restaurant locations coloured by rating

Correlation with rating: latitude r = -0.12, longitude r = -0.21. Weak overall, because location acts through country/city clusters.

Country	Rated restaurants	Avg rating
Philippines	22	4.47
Turkey	34	4.30
Indonesia	21	4.30
New Zealand	40	4.26
UAE	60	4.23
South Africa	60	4.21
United Kingdom	79	4.14
Brazil	55	4.11
Qatar	20	4.06
United States	431	4.03
Sri Lanka	20	3.87
Australia	24	3.66
Canada	4	3.58
Singapore	20	3.58
India	6513	3.35

India has the lowest average rating among rated restaurants (3.35) but by far the most restaurants; the Philippines, Indonesia and Turkey rate highest.

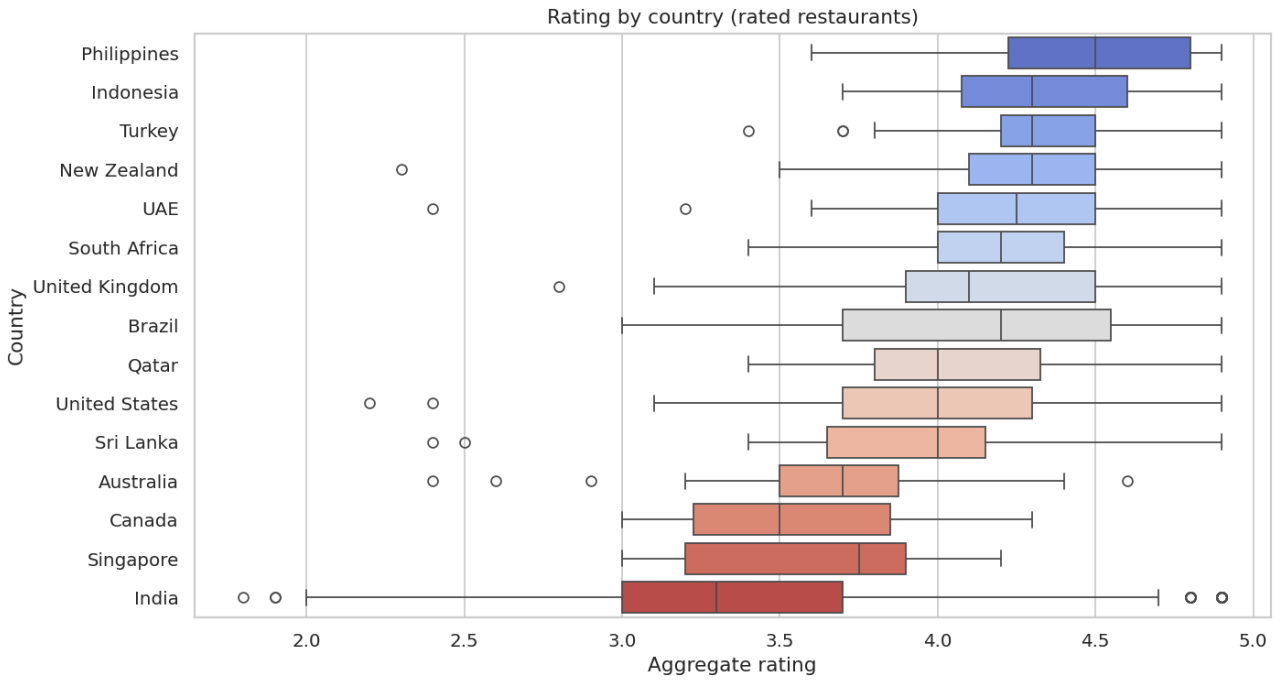


Figure 4 - Rating spread by country

Level 2

Task 1 – Table Booking & Online Delivery

- Table booking available: **12.12%**; online delivery available: **25.66%**.
- Average rating with table booking **3.44** vs without **2.56** (rated restaurants only: 3.59 vs 3.41, so part of the gap is due to unrated places).

Online delivery by price range

Price range	No delivery %	Online delivery %
1.00	84.20	15.80
2.00	58.70	41.30
3.00	70.80	29.20
4.00	91.00	9.00

Delivery is most common in price range 2 (41.3%) and rarest in range 4 (9.0%), since fine dining tends to be dine-in.

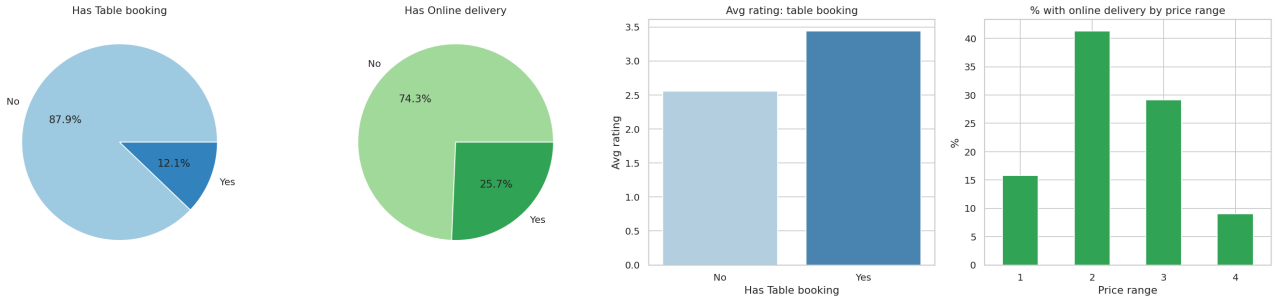


Figure 5 - Table booking, online delivery and ratings

Task 2 – Price Range Analysis

Price range	Restaurants	Share %	Avg rating
1.00	4444.00	46.50	2.00
2.00	3113.00	32.60	2.94
3.00	1408.00	14.70	3.68
4.00	586.00	6.10	3.82

Most common price range: 1 (4,444 restaurants). **Colour with the highest average rating:** Dark Green (4.66).

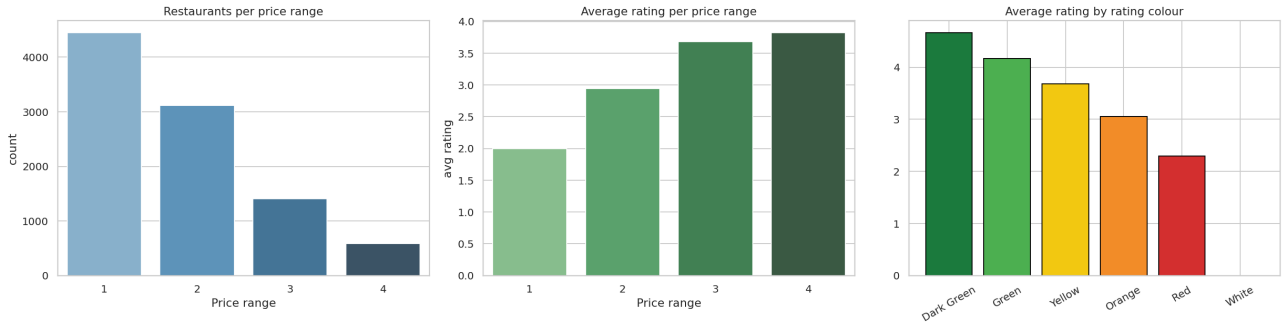


Figure 6 - Price range counts, average rating and rating colour

Task 3 – Feature Engineering

- **Length features:** Name Length, Name Word Count, Address Length, Address Word Count, Locality Length.
- **Cuisine features:** Cuisine Count, Primary Cuisine.
- **Encoded flags:** Has Table Booking, Has Online Delivery, Is Delivering Now (Yes/No to 1/0).
- **Other:** Cost per Person, Is Rated.

Feature	Correlation with rating
Name Length	-0.035
Address Length	0.002
Cuisine Count	0.248
Has Table Booking	0.190
Has Online Delivery	0.226
Votes	0.314

Cuisine count, delivery and table booking carry signal; name and address length are essentially unrelated to rating.

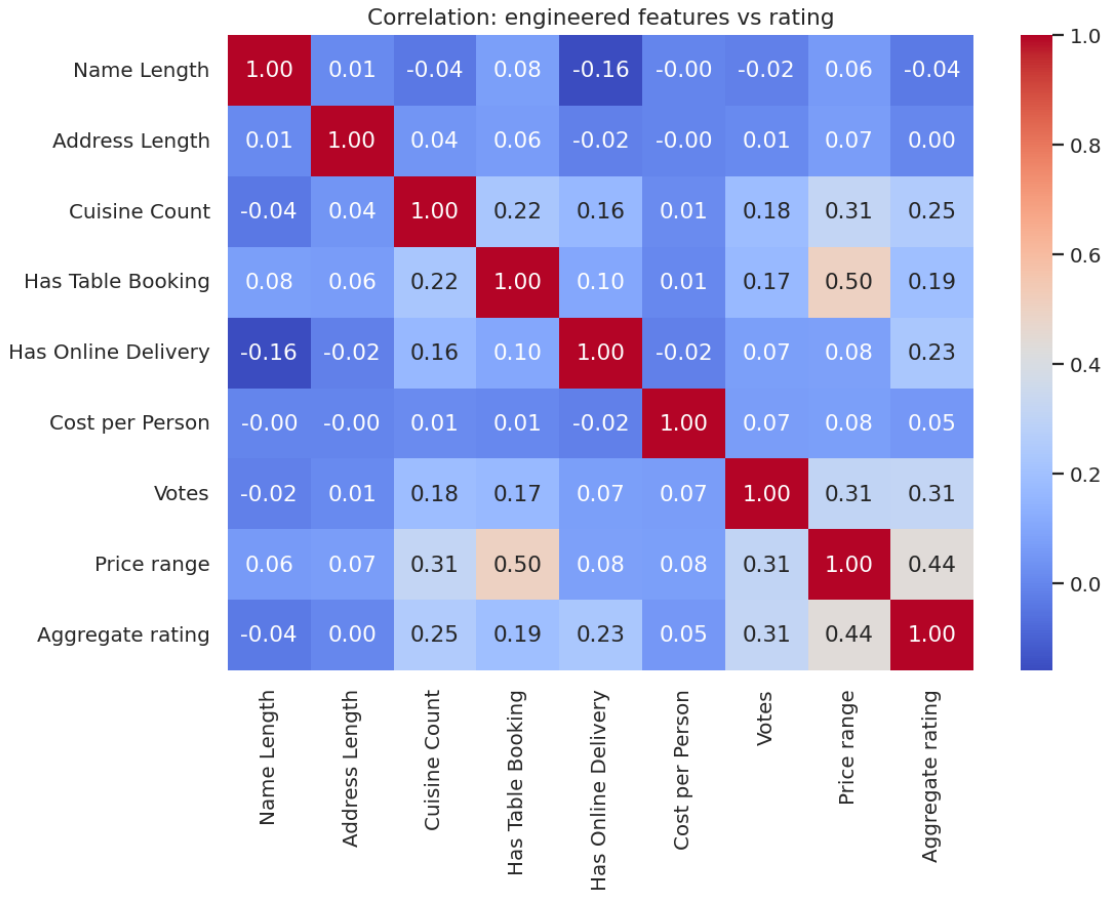


Figure 7 - Correlation heatmap of engineered features

Level 3

Task 1 – Predictive Modeling

Target: Aggregate rating. Features: country, city frequency, latitude/longitude, log cost, price range, votes, booking/delivery flags, cuisine count, primary-cuisine frequency, name and address length. 80/20 train-test split (random_state=42), plus 5-fold cross-validation.

Run A - all restaurants

Model	MAE	RMSE	R2 (test)	R2 (train)	R2 (5-fold CV mean)
Random Forest	0.192	0.292	0.962	0.991	0.947
Gradient Boosting	0.199	0.294	0.962	0.969	0.949
Decision Tree	0.216	0.325	0.954	0.961	0.940
Linear Regression	1.045	1.252	0.312	0.325	0.212
Ridge Regression	1.045	1.252	0.312	0.325	0.212

Run B - rated restaurants only (realistic)

Model	MAE	RMSE	R2 (test)	R2 (train)	R2 (5-fold CV mean)
Gradient Boosting	0.244	0.327	0.655	0.713	0.487
Random Forest	0.245	0.330	0.648	0.913	0.488
Decision Tree	0.267	0.362	0.577	0.649	0.401
Linear Regression	0.334	0.425	0.416	0.393	0.299
Ridge Regression	0.334	0.425	0.416	0.393	0.299

Important: Run A reaches $R^2 \approx 0.96$ only because unrated restaurants have 0 votes and 0 rating (the model learns 'no votes = no rating'). Run B removes this and gives a fair estimate: tree ensembles (Gradient Boosting / Random Forest) clearly beat linear models (R^2 0.65 vs 0.42), showing non-linear relationships. Random Forest overfits more (train R^2 0.91 vs test 0.65).

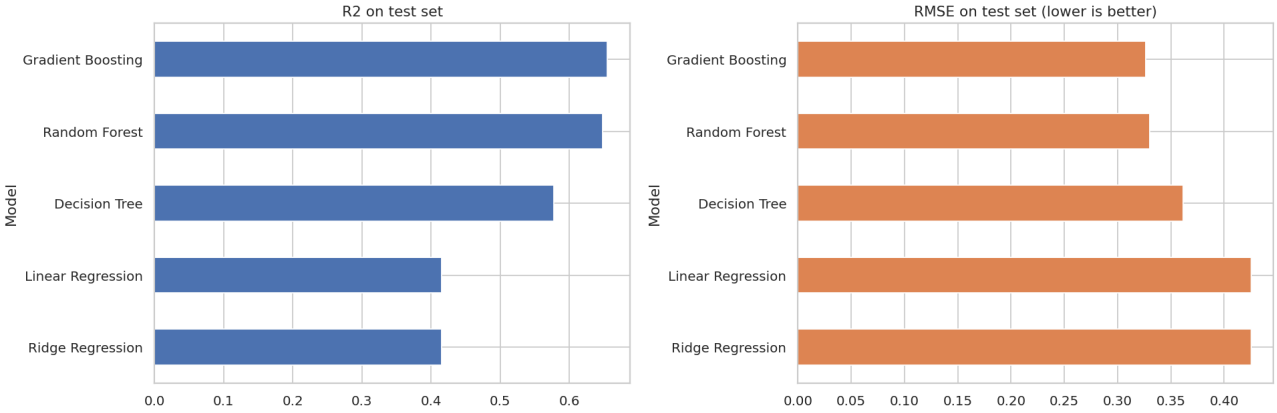


Figure 8 - Model comparison (rated only)

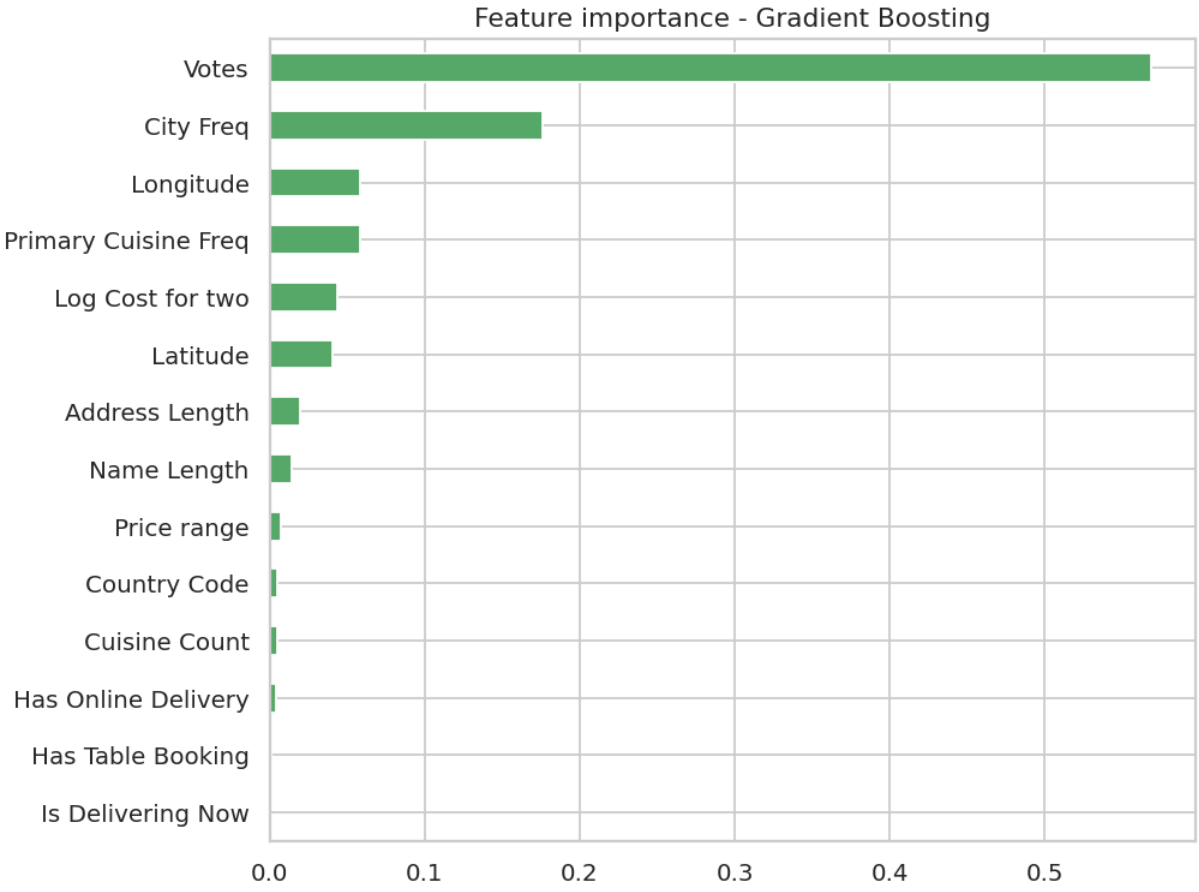


Figure 9 - Feature importance (rated only)

Task 2 – Customer Preference Analysis (Cuisines)

Most popular cuisines (by total votes)

Cuisine	total_votes	restaurants	avg_rating
North Indian	595981	3960	3.30
Chinese	364351	2735	3.28
Italian	329265	764	3.75
Continental	288255	736	3.71
Fast Food	184058	1986	3.26
American	183117	390	3.76
Cafe	177568	703	3.68
Mughlai	151946	995	3.27

Highest-rated cuisines (at least 20 restaurants)

Cuisine	avg_rating	restaurants	total_votes
Brazilian	4.34	28	3259
International	4.25	21	8122
Indian	4.16	70	22396
Bar Food	4.15	39	15684
Southern	4.13	24	13939
Vegetarian	4.07	23	10714
Sandwich	4.07	53	23500
Grill	4.06	21	4301

Mainstream cuisines (North Indian, Chinese, Fast Food) dominate votes but rate below the overall mean of 3.47, while niche or international cuisines (Brazilian, International, Bar Food, Mediterranean) rate higher. A Kruskal-Wallis test confirms cuisine ratings differ significantly.

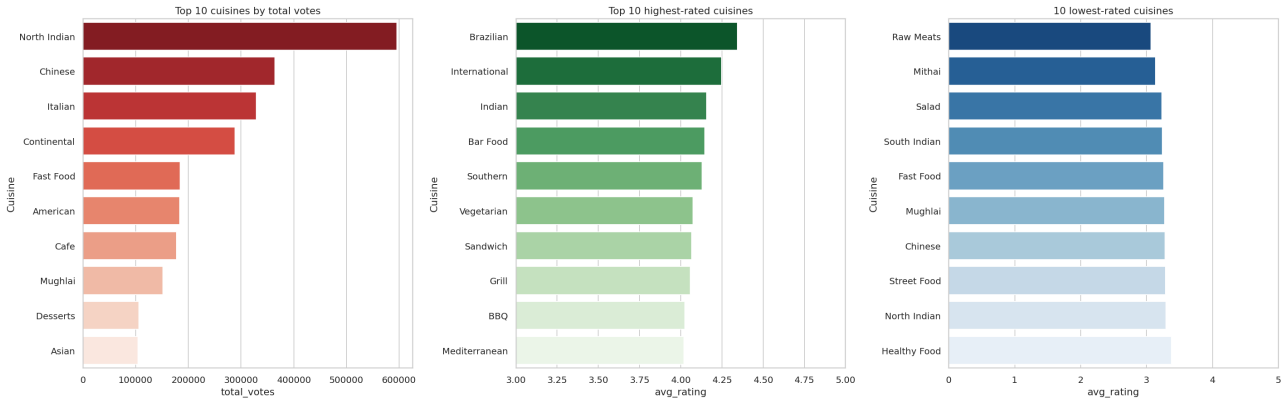


Figure 10 - Popular, top-rated and lowest-rated cuisines

Task 3 – Data Visualization

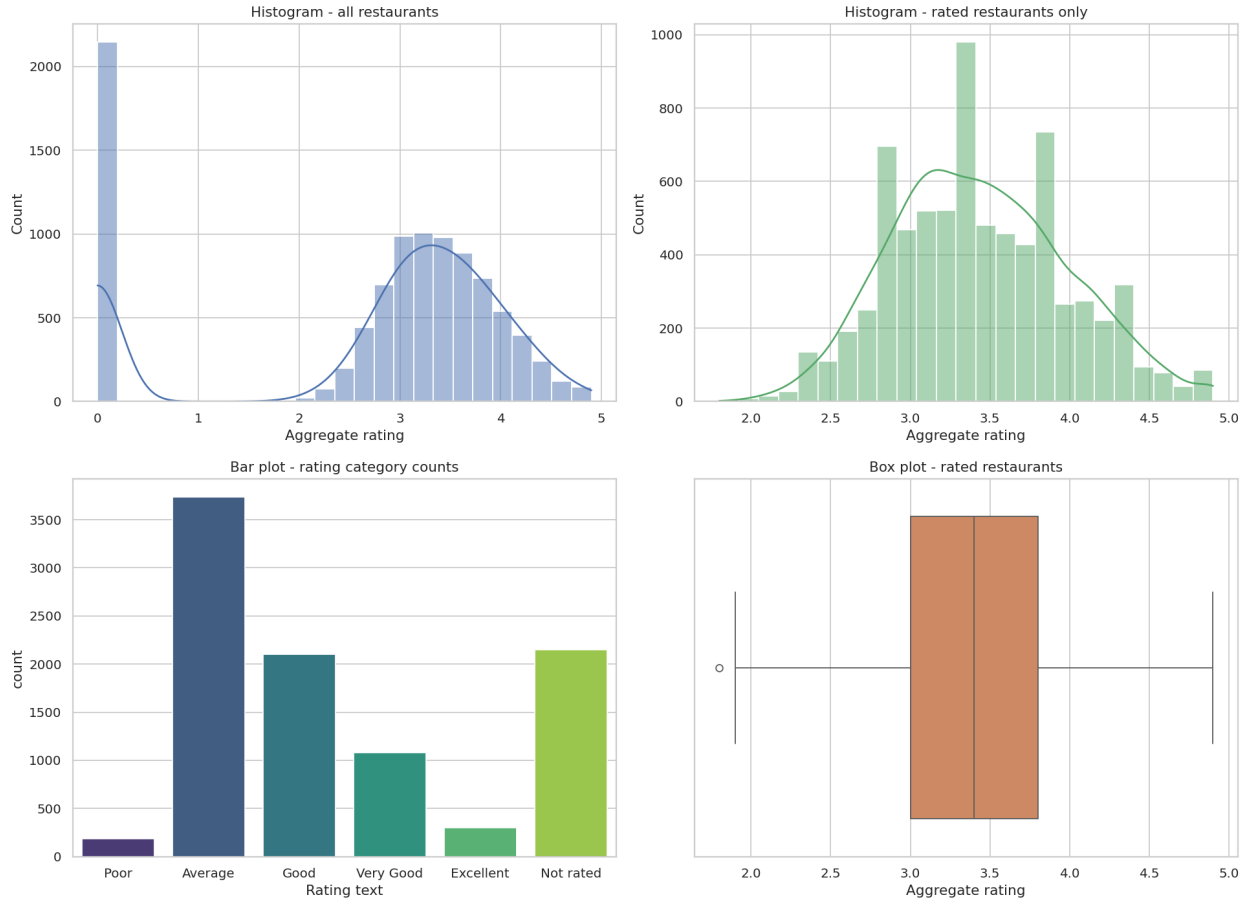


Figure 11 - Distribution of ratings (histogram, bar plot, box plot)

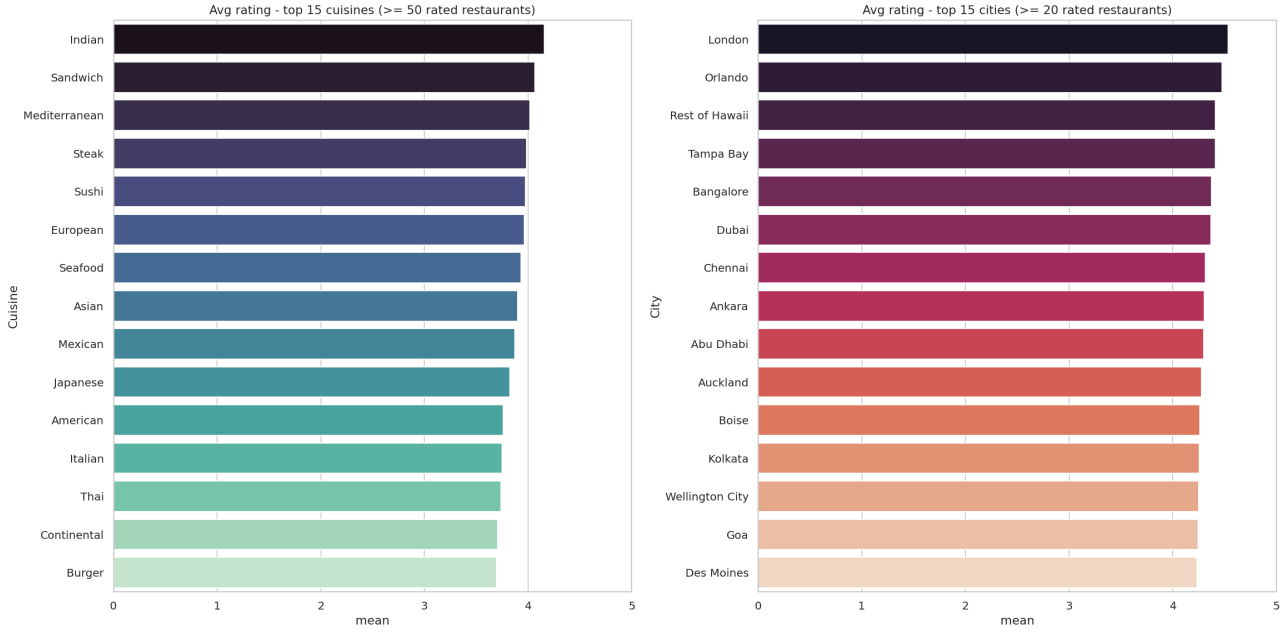


Figure 12 - Average rating of top cuisines and cities

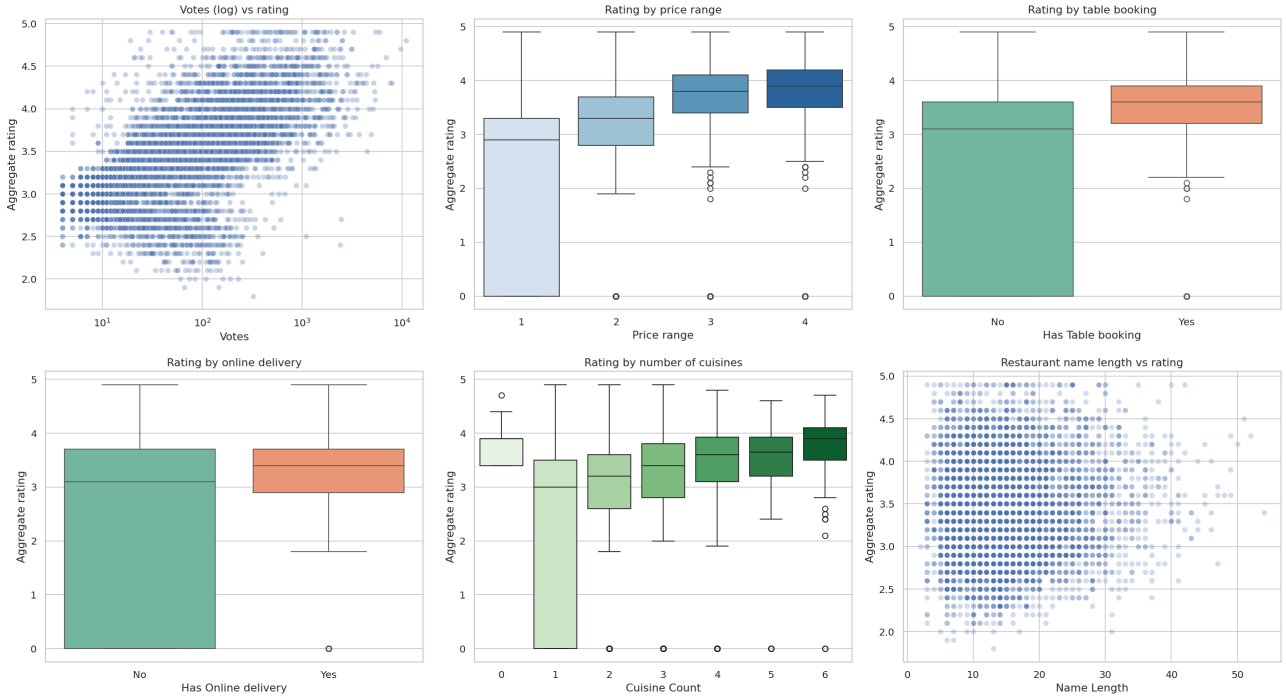


Figure 13 - Relationship between features and rating

Key Insights

- Price range and votes have the strongest links with rating; popular and pricier restaurants rate higher.
- Table booking and online delivery are associated with higher ratings.
- Name length, address length and raw coordinates have almost no relationship with rating.
- Unrated (0) restaurants should be filtered out when comparing averages or training models.